

ABDUL MANNAN

AI / ML Engineer

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SUMMARY

AI/ML engineer who ships LLM, RAG, and computer-vision systems to production - and builds the backend that serves them. Delivered RAG assistants and document-intelligence pipelines handling 5,000+ inference requests/day over a 10,000+ document knowledge base, plus the async FastAPI/Node services, monitoring, and CI/CD around them. Strong at turning model prototypes into reliable, evaluated, low-latency APIs.

TECHNICAL SKILLS

AI / ML: LLMs, RAG, LangChain, Hugging Face, OpenAI / GPT-4o Vision, embeddings & vector search (pgvector), prompt engineering, model evaluation, Computer Vision, OCR, PyTorch, TensorFlow, OpenCV

Languages: Python, TypeScript, JavaScript (ES6), SQL

Backend & Serving: FastAPI, Flask, Node.js, NestJS, Express, REST APIs, WebSockets (Socket.IO)

Data & Infra: PostgreSQL (Prisma ORM), pgvector, Docker, AWS S3, Firebase, Google Cloud Storage

Practices: Async services, unit & e2e testing (Jest / pytest), CI/CD (GitHub Actions), modular architecture, code review

WORK EXPERIENCE

AI Engineer - SingX Solutions (subsidiary of Devtects)

Karachi, Pakistan | Aug 2025 - Present

- Designed 10 RAG assistants (LangChain, Hugging Face) delivering document-grounded Q&A over a 10,000+ document knowledge base for internal and customer-facing users.
- Built OCR and document-intelligence pipelines (RapidOCR, OpenCV, PyTorch, GPT-4o Vision) that auto-extract structured fields from identity documents in ~1.3s - down from ~3 minutes of manual entry.
- Shipped ML/LLM features as async FastAPI/Flask microservices serving 5,000+ real-time inference requests/day, monitored for uptime and latency and released via GitHub Actions CI/CD.

Full Stack Engineer - Devtects

Karachi, Pakistan | Aug 2024 - Aug 2025

- 1stBond (social + marketplace):** Engineered a modular NestJS backend (19+ modules, PostgreSQL/Prisma, Socket.IO) for real-time chat, location sharing, and panic alerts, with Jest unit and e2e coverage.
- AssanPay (fintech payments):** Built the core backend (Node.js, TypeScript, Express, PostgreSQL/Prisma) for merchant onboarding, payments, and settlements - auth, KYC, wallet ledger, and QR payments.

Freelance AI Developer - Remote

2022 - Present

- Delivered end-to-end computer-vision, NLP, and generative-AI projects, taking models from prototype to deployed API (Python, FastAPI, Flask, Streamlit), including LLM and Telegram-bot integrations for external clients.

PROJECTS

- CNIC OCR & Document Extraction** - OCR pipeline (RapidOCR, OpenCV, GPT-4o Vision) extracting structured fields from Pakistani CNICs via a Flask API, with custom preprocessing for low-quality scans.
- EclatAI - AI-vs-Real Image Classifier** - ResNet50 + SVM with a Streamlit dashboard and SHAP explainability that shows why an image was flagged AI-generated.
- AgentForge - Multi-tenant RAG Platform (in progress)** - Config-driven AI agent platform: FastAPI, PostgreSQL + pgvector, Gemini embeddings, tenant-isolated document Q&A with [n] citations.
- Human Pose Estimation** - YOLO-based real-time human keypoint detection for live video input.

EDUCATION

BSc in Artificial Intelligence - Sindh Madressa-tul-Islam University, Karachi | Sep 2020 - Sep 2024